

Elias Weston-Farber

Research Engineer — Scientific AI & Clinical Trial Data Infrastructure

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Research engineer building AI systems for scientific evidence synthesis and clinical trial data infrastructure. Created ScienceAI, an open-source agentic harness that coordinates frontier language models for systematic literature analysis with provenance tracking. 9 peer-reviewed publications including NEJM and JAMA. BSPH AI Research Day Staff Award, 2026.

Publications & Awards

9 peer-reviewed publications including **NEJM** and **JAMA** — 2 as author; 7 as a named METRC consortium collaborator, credited for designing and operating the clinical trial data pipelines, quality systems, and DSMB reporting infrastructure underlying multi-center randomized trials, including the 12,211-patient PREVENT CLOT trial (NEJM 2023) and the TOBRA trial (JAMA 2026) • **4 manuscripts in press / under review** (METRC) • **1 DOI-archived open-source software package** ([10.5281/zenodo.21938772](https://doi.org/10.5281/zenodo.21938772))

Full publication list: [PubMed Bibliography](#)

BSPH AI Research Day Staff Award — Johns Hopkins Bloomberg School of Public Health, March 2026. Awarded for podium presentation and poster session on ScienceAI alongside Jeff Leek's keynote. Selected from submitted abstracts across BSPH. Abstract co-authors: Renan Castillo, Katherine Frey.

Selected Projects

ScienceAI — Open-Source Agentic Harness for Scientific Literature Analysis (2024–present) - Designed and built a multi-agent orchestration system for systematic review and meta-analysis tasks: PI orchestrator delegates to Analyst Agents that perform structured data extraction with reflection/validation - Supports GPT, Claude, and Gemini as interchangeable backends with unified provider abstraction; field-level provenance tracking links every extracted data point to source quotes, page locations, and derivation explanations - Validated by reproducing Scolaro et al. (2014) meta-analysis: three frontier models independently recovered the primary finding (OR 2.15–2.37 vs. published 2.32) from 19 source papers - Published on PyPI (`pip install scienceai-llm`), GPL-3.0, 18 releases, full CI/CD, archived on Zenodo

Baltimore Comedy Place — Fully autonomous, AI-powered data pipeline that scrapes, cleans, and aggregates event listings from disparate sources to create Baltimore's first comprehensive comedy show guide

EliasTechLabs — Serverless personal site built on AWS Lambda, CloudFront, and S3, featuring an LLM-powered website assistant

RSmartsheet — R package interfacing with Smartsheet API, used by health authorities for project management

SearchIt — Systematic web search application used by NGOs for public data gathering and analysis

Cloud Platforms — Azure, GCP, AWS

Production experience across all three major clouds: **Azure** (current CAS platform), **GCP** (prior CAS platform, 2019–2026), **AWS** (serverless Lambda/CloudFront/S3 deployments).

CAS Platform Migration, GCP → Azure (2026) — architect and lead engineer - Migrated METRC's multi-study data and reporting platform (26 TB) from GCP to Azure with a zero-downtime DNS cutover: App Engine → Container Apps, Compute Engine → Container Instances (on-demand 16-vCPU/64 GB self-stopping worker), Firebase → Azure SQL, Cloud Storage → Blob Storage, Google Workspace → SharePoint via Microsoft Graph - Replaced Google OAuth with institutional single sign-on (Shibboleth OIDC), including an MSAL token-delegation workaround for enterprise RBAC constraints - Hardened to institutional compliance standards: secretless container images via Key Vault and managed identities, VNet-injected apps with private endpoints, SQL firewall reduced from 94 rules to 2, centralized audit logging with 365-day retention and alerting - Captured the full production footprint (33 resources) in Terraform with zero-drift verification and a terraform-first change policy - Authored the disaster-recovery/business-continuity plan, PHI compliance plan, and institutional IT risk documentation for security review

Work Experience

Senior Cloud Engineer & Data Manager

METRC, Johns Hopkins Bloomberg School of Public Health 2023 – Present

- Manage cloud infrastructure and data pipelines powering multi-site randomized controlled trials across 70+ Level I trauma centers nationwide
- Engineered an automated DSMB reporting system for nine studies, replacing a manual approach that required five full-time analysts with one that now requires only a single analyst
- Decreased cloud costs by 20% by migrating parts of the backend to Compute Engine while simultaneously improving system responsiveness and user engagement through HTMX and WebSockets, enabling real-time feedback
- Implemented a real-time Firebase database to ensure data synchronization across distributed environments

Senior Research Application Developer & Data Manager

Johns Hopkins Bloomberg School of Public Health 2020 – 2023

- Led and managed a team of six analysts and developers, successfully setting up seven studies on an analytic platform built on GCP, automating weekly reporting
- Designed and implemented a system for real-time study data issue tracking and notifications, replacing a manual static file-based system and facilitating rapid response measures that supported a major publication
- Developed tailored visualization and reporting solutions for research teams, integrating technologies such as Selenium and Scrapy to extract conflicts of interest data, and Plotly to create interactive reports

Research Application Developer

Johns Hopkins Bloomberg School of Public Health 2019 – 2020

- Developed a robust administrative data storage solution for managing records across 30+ METRC studies, leveraging custom Smartsheet & Google Sheets APIs and a cross-platform Electron application
- Wrote multiple R packages and designed a Python+R Docker environment to standardize the computational environment for METRC study analysis

Backend Developer & Data Scientist

Civicly Involved, Chicago 2018 – 2019

- Led a team of interns in designing and implementing API-based data integration and management solutions, including geospatial data analysis using Google Cloud technologies

Education

University of Maryland Baltimore County, Baltimore — 2018 *Bachelor of Science in Environmental Science & Geography Bachelor of Arts in Political Science*

Technical Skills

- **Languages:** Python, R, JavaScript, SQL
- **AI/ML:** Multi-agent LLM orchestration, provider API integration (OpenAI, Anthropic, Google), structured data extraction, prompt engineering, evaluation design
- **Frameworks:** Flask, Node.js, PyTorch, Pandas, Tidyverse, Shiny, Electron, HTMX
- **Infrastructure:** Docker, Azure (Container Apps, Azure SQL, Key Vault, VNet), GCP, AWS, Terraform, GitHub Actions, Firebase, WebSockets
- **Data & Analysis:** Tableau, ArcGIS, QGIS, Plotly, ggplot2